Abdelrahman Khalifa

Software Engineer | Python, Rust, SQL, Linux

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SUMMARY

Software engineer who programs for fun as much as for work: 11 PRs merged upstream as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), plus merged changes in Zephyr RTOS, Rust Clippy and the Boa JavaScript engine. Writes Python, Rust, C and C++; uses Linux every day. Teaching himself database internals through Carnegie Mellon's database course. Codeforces Expert. Used to remote, asynchronous work and written communication.

EXPERIENCE

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | May 2026 – Oct 2026

Remote open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | *github.com/Abd002/GSOC-2026*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI, working remotely with mentors from OpenPrinting and System76 (cosmic-settings #2182, in review).
- Designed it as 5 Rust crates with an async client, and ran the C FFI backend as a separate varlink IPC service so a crash in native code cannot take down Settings.
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with a CI matrix across architectures and CUPS versions, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer | **EKSON** | Aug 2025 – Present

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Developed the Python (PyQt5) control app; used asyncio to wait for ESP32 and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Built remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware; GitHub Actions CI.
- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | Dec 2024 – Jan 2025

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

Contributor (all merged) — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

PROJECTS

- **Database Systems, CMU 15-445** (C++17, SQL): thread-safe Count-Min Sketch for the BusTub database (per-row locks, merge, top-k); SQL analytics in DuckDB (joins, aggregation, upserts). GitHub
- **Kernel Monitor** (C, Linux kernel): character device driver that exports per-task memory, threads, context switches and CPU time to user space using RCU. GitHub
- **Function Solver and Plotter** (Python, PySide2, SymPy, Matplotlib): solves and plots two functions with their intersections; input validation and end-to-end tests with pytest-qt. GitHub

TECHNICAL SKILLS

Languages: Python (asyncio), Rust, C, Modern C++, SQL, JavaScript (Node.js), Bash

Data & backend: SQL (DuckDB, SQL Server), MongoDB, REST APIs, Express; database internals (storage, buffer pool, hash tables)

Linux & tools: Linux daily driver (Arch-based CachyOS), kernel modules, Git/GitHub, GitHub Actions CI, Cargo, CMake, pytest, debugging, code review

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024; mentor in the local ICPC community.

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | 2020 – 2025

- Grade: Very Good with Honors.